

Day 13 Activity: Reshaping data

Thursday, June 4, 2026

Your Name Here

Setup

```
library(tidyverse)
library(palmerpenguins)
```

Today's verbs are `pivot_longer()` and `pivot_wider()`. We'll use a few small built-in tables plus penguins.

Part 1 — Tidy, wide, or narrow?

Look at `table1` and `table2` (both built into `tidyr`):

```
table1
```

```
# A tibble: 6 x 4
  country    year cases population
  <chr>      <dbl> <dbl>      <dbl>
1 Afghanistan 1999    745  19987071
2 Afghanistan 2000   2666  20595360
3 Brazil      1999  37737  172006362
4 Brazil      2000  80488  174504898
5 China       1999 212258 1272915272
6 China       2000 213766 1280428583
```

```
table2
```

```
# A tibble: 12 x 4
  country    year type      count
  <chr>    <dbl> <chr>    <dbl>
1 Afghanistan 1999 cases      745
2 Afghanistan 1999 population 19987071
3 Afghanistan 2000 cases     2666
4 Afghanistan 2000 population 20595360
5 Brazil      1999 cases     37737
6 Brazil      1999 population 172006362
7 Brazil      2000 cases     80488
8 Brazil      2000 population 174504898
9 China       1999 cases     212258
10 China      1999 population 1272915272
11 China      2000 cases     213766
12 China      2000 population 1280428583
```

1a. For each, describe what **one row** represents.

[table1: ...] — [table2: ...]

1b. Which one is **tidy** (each variable a column, each observation a row)? Which is the **narrower** of the two, and how can you tell?

[Your answer.]

Part 2 — `pivot_longer()` on a tiny table

Build this wide table by hand:

```
bp_wide <- tribble(
  ~subject, ~before, ~after,
  "BHO",    160,    160,
  "GWB",    115,    135,
  "WJC",    105,    145
)
bp_wide
```

```
# A tibble: 3 x 3
  subject before after
  <chr>    <dbl> <dbl>
1 BHO         160   160
2 GWB         115   135
3 WJC         105   145
```

2a. Predict first. If you pivot `before` and `after` into a long form, how many **rows** will the result have? How many **columns**?

Prediction: [... rows / ... columns]

2b. Now do it. Use `names_to = "when"` and `values_to = "sbp"`. Check against your prediction.

```
# Your code:
```

Part 3 — `pivot_longer()` on real data

The billboard data (in `tidyr`) has one column per chart week (`wk1`, `wk2`, ...).

3a. Pivot all the `wk` columns into `week` / `rank`, dropping the empty weeks. (Hint: `cols = starts_with("wk")`, `values_drop_na = TRUE`.)

```
# Your code:
```

3b. Add `mutate(week = parse_number(week))` so `"wk3"` becomes `3`, then plot `rank` (y) against `week` (x), one faint line per track. Use `scale_y_reverse()` so rank 1 sits on top.

```
# Your code:
```

One sentence: what does the plot show about how long songs stay on the charts?

[Your answer.]

Part 4 — `pivot_wider()` for comparison

Start from a **long** summary (this is the kind of table you built on Wednesday):

```
penguin_summary <- penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(avg_mass = mean(body_mass_g, na.rm = TRUE), .groups = "drop")
penguin_summary
```

```
# A tibble: 6 x 3
  species sex   avg_mass
  <fct>   <fct>   <dbl>
1 Adelie female  3369.
2 Adelie male   4043.
3 Chinstrap female 3527.
4 Chinstrap male   3939.
5 Gentoo female  4680.
6 Gentoo male   5485.
```

4a. Predict first. If you widen this with `names_from = sex`, `values_from = avg_mass`, how many rows and columns will you get?

Prediction: [... rows / ... columns]

4b. Do the pivot, then add a column `diff = male - female`.

```
# Your code:
```

Which species has the biggest mass gap between sexes?

[Your answer.]

Part 5 — The round trip, and a trap

5a. Take your widened penguin table from Part 4 (before the `diff` column) and pivot it **back** to long form. Do you recover the original `penguin_summary`? (`pivot_longer()` and `pivot_wider()` are inverses.)

```
# Your code:
```

5b. Debug it. Someone tries to make `bp_wide` longer like this and gets a strange result:

```
bp_wide |>
  pivot_longer(
    cols = c(subject, before, after),
    names_to = "when",
    values_to = "sbp"
  )
```

Run it. Why is this “too narrow” / wrong? What single change to `cols` fixes it?

[Your explanation and fix.]

Part 6 — Choose the shape

For each goal, say whether you’d want the data **wide** or **narrow**, and which pivot gets you there. One sentence each.

1. Plot each country’s TB cases over time, one line per country.
2. Compute, for each country, the change in cases from 1999 to 2000 as a single new column.
3. Group by year and sum the cases across all countries.

[1: ...] — [2: ...] — [3: ...]

Wrap-up

One-sentence reflection: pivoting is the first thing this week that changed the *shape* of the table rather than its contents, what clicked, and what’s still fuzzy?

[Write one sentence.]

Code appendix

```
library(tidyverse)
library(palmerpenguins)
table1
table2
bp_wide <- tribble(
  ~subject, ~before, ~after,
  "BHO",      160,    160,
  "GWB",      115,    135,
  "WJC",      105,    145
)
bp_wide
# Your code:

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penguin_summary <- penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(avg_mass = mean(body_mass_g, na.rm = TRUE), .groups = "drop")
penguin_summary
# Your code:

# Your code:

bp_wide |>
  pivot_longer(
    cols = c(subject, before, after),
    names_to = "when",
    values_to = "sbp"
  )
```